

Supplementary A — Extended motivation for the band-mask robustness diagnostic

Companion to: *A Band-Mask Robustness Diagnostic for LDA on HSI* (WHISPERS 2026)

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A.1 Scope

This supplement expands Section I of the conference manuscript. The main paper limits itself to a single contribution — a band-mask robustness diagnostic for LDA on HSI — presented in four printed pages. Here we provide (a) the historical context that motivated the diagnostic, (b) a complete enumeration of the failure modes the diagnostic intends to surface, (c) a contrast with adjacent diagnostics in the literature, and (d) the threats to validity we acknowledge upfront.

B.2 Historical context

Latent Dirichlet Allocation [1] introduced the generative-topic-model framing of a discrete-token corpus over a decade before its first reported application to hyperspectral imagery. The application to HSI [2, 3] was preceded by two adjacent paradigms whose evaluation methodology shaped the field’s expectations: linear spectral unmixing [4, 5] and deep spectral-spatial classification [6, 7]. Both paradigms matured under an *accuracy-first* evaluation discipline: the canonical figure of merit is overall accuracy (OA) of a downstream classifier on a held-out fold of one of the standard labelled benchmarks (Indian Pines, Salinas, Pavia U; later KSC and Botswana), and methodological progress is measured as a delta against the best prior OA on the same scene.

This discipline served unmixing and deep classification well: the inductive bias of each paradigm is sufficiently strong that post-hoc interpretability claims (endmember spectra; feature visualisations) carry low risk of being a freak of a particular training run. Topic models do not enjoy that protection. The basis $\phi \in \mathbb{R}_+^{K \times B}$ inferred by LDA is identifiable only up to a label permutation, and the inference procedure (online variational Bayes [8], collapsed Gibbs sampling [9], or amortised variational autoencoders in the neural variants [10, 11]) has multiple posterior modes for any finite-data problem. The interpretability of a single trained model is therefore contingent on a chain of stability properties:

1. **Seed stability:** refitting under a different random initialisation should yield the same basis up to permutation.
2. **Capacity stability:** increasing K by one should split one canonical topic into two, not reshuffle every topic.
3. **Sample stability:** subsampling the corpus should not relabel topics.
4. **Hyperparameter stability:** nudging α or η should not alter topic identity at the same K .
5. **Spectral-window stability:** dropping the atmospheric-water bands, or restricting to VNIR or SWIR only, should not relabel topics (modulo the bands that the mask actually removes).

Failure of any one of these properties weakens any claim of the form “topic k is a mineral signature” or “topic k corresponds to healthy canopy”. The first four properties are reported (sparingly) in the existing literature; the fifth — spectral-window stability — is, to our knowledge, not reported anywhere as a per-scene diagnostic. The band-mask diagnostic of the main paper fills exactly this gap.

C.3 Failure modes the diagnostic surfaces

The band-mask diagnostic is a hypothesis test under the null: H_0 : *The canonical topic basis is a property of the scene’s underlying material distribution and is therefore robust to the choice of spectral window.*

The diagnostic returns a per-(scene, mask) paired ARI; values close to 1 fail to reject H_0 , values close to 0 reject. There are three substantively different ways to reject:

- **Mode (i): canonical-fit redundancy.** The canonical basis exploits fine-grained correlations that are present in the full-band representation and disappear under restriction. The canonical topics are correct in the canonical reference frame but not transportable across spectral windows.
- **Mode (ii): weak identification.** The latent basis is under-determined by the available data (small per-class sample, large class count, narrow bands). Different posterior modes are equally plausible under different band subsets.
- **Mode (iii): mask-induced collinearity.** The mask removes the bands that were carrying the discriminative signal, so the masked LDA fits a different basis on the residual spectrum that is correct for the residual but incomparable to the canonical.

The diagnostic does not separate (i)–(iii) by design — doing so would require additional information (e.g. band-wise Fisher ratios, sample-size posterior bounds) and would inflate the diagnostic’s artefact footprint without changing the per-scene yes/no judgement that motivates it. The conference paper’s discussion explicitly flags that the diagnostic is a *flag-raising* tool, not a *root-cause-isolating* tool.

D.4 Contrast with adjacent diagnostics

We briefly contrast the band-mask diagnostic with three related diagnostics in the wider topic-model and clustering literature.

Seed-stability swarms. Plotting per-seed ARI to ground truth across N refits [12, 13] answers “does the same model on the same data yield the same topics”, a different and weaker question than “does the same model on related data yield the same topics”. Seed stability is, in our framework, axis F-3; the band-mask diagnostic is axis F-5.

Cross-fold ARI. K-fold CV on θ followed by ARI comparison is sensitive to the held-out fold’s class proportion; it does not surface whether the topic basis itself is stable.

Coherence measures. c_v and c_{NPMI} [14] score the top- N words per topic against a reference corpus; they capture intra-topic compactness but say nothing about cross-fit identity preservation.

The band-mask diagnostic occupies a different methodological niche from each of the above: it asks whether the inferred topic *identities* survive a specific class of data restriction, with the answer interpretable directly as a chance-corrected proportion.

E.5 Threats to validity acknowledged upfront

Top-50 Fisher ranking is computed on the unmasked spectrum. The Fisher ratio per band is computed against the ground-truth label on the full-band spectrum; the mask then selects the top-50 bands by that ranking. This is a deliberate choice — recomputing the Fisher ranking on each masked subset would conflate the within-mask discriminability with the between-mask comparison the diagnostic is trying to surface. The trade-off is that the top-50 Fisher mask is a *post-hoc* selector and its paired-ARI value should be interpreted accordingly.

Canonical K is held fixed. The mask-refit uses the canonical K rather than a per-mask K^* . We argue this is the right comparison because changing K between fits introduces a second degree of freedom that mixes with the band-restriction we are isolating. The cost is that on a scene where the canonical K is too large for the masked spectrum, the masked fit will under-utilise its capacity.

Sentinel masking of unlabelled pixels. Both the canonical and masked dominant-topic maps use sentinel 255 for unlabelled pixels; the paired ARI is computed only on the intersection of labelled pixels. The two fits therefore see a different document set from the band axis but the same document set from the spatial axis.

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